

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26-11-2025
Name	Yakshagaraj A G
Project Name	Online Bookstore
Maximum Marks	4 Marks

Technical Architecture:

The Online Bookstore platform follows a robust and scalable 3-tier architecture, ensuring a smooth user experience for browsing, searching, and purchasing books.

Presentation Layer (Frontend)

- User-friendly interface for browsing, searching, and purchasing books
- Responsive design for desktop and mobile
- Provides separate admin UI for book management

Application Layer (Backend / Business Logic)

- Handles user authentication, book search, cart operations, orders, and admin controls
- REST APIs for the frontend
- Secure processing of order data

Data Layer (Database)

- Stores users, books, categories, cart items, orders, and reviews
- Ensures consistent and safe storage of all platform data

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web-based interface for users/admin	React.js, HTML, CSS, JavaScript, Tailwind CSS
2	Application Logic – 1	Book search, filters, cart handling	Node.js, Express.js
3	Application Logic – 2	Admin panel, order management, reviews	React.js + Node.js
4	Database	Stores books, users, orders, reviews	MongoDB (Atlas)
5	Authentication	Secure login and registration	JWT, bcrypt.js
6	API Communication	Frontend-backend communication	REST API, Axios
7	Deployment	Hosting & CI/CD	Vercel / Netlify / Render / Railway

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Modern frontend & backend tools	React.js, Node.js, Tailwind CSS
2	Scalable Architecture	Supports large inventory & many users	MVC + REST API
3	Secure Framework	Role-based access for admin & users	JWT, bcrypt
4	Responsive UI	Works smoothly on phones & desktops	React.js + Tailwind
5	Cloud Storage	Image storage for book covers	Cloudinary (optional)

References:

[React.js Documentation](#)

[Node.js Best Practice](#)

[JSON Web Server Reference](#)